



ArcGIS Pro Track — Professional Cartography & Analysis

GeoCascade Chapter 1 — ArcGIS Pro Workflow Guide *ArcGIS Pro 3.x + arcpy | Torres del Paine, Patagonia*

Overview

This track uses ArcGIS Pro to create professional cartographic outputs from the data produced by the Python pipeline in Chapter 1. All Python-generated GeoTIFFs, CSVs, and GeoPackages are natively compatible with ArcGIS Pro.

Script	What it does	Run how
01_arcpy_import_imagery.py	Mosaic DEM, composite S2, import all layers	ArcGIS Pro Notebook
02_arcpy_climate_maps.py	Apply professional symbology to all layers	ArcGIS Pro Notebook
03_arcpy_layout_export.py	Build 4-panel layout, export PDF + PNG	ArcGIS Pro Notebook

Step 1: Set Up the ArcGIS Pro Project

Create a new project

1. Open **ArcGIS Pro 3.x**
2. **New > Map** → name the project **GeoCascade_Ch01**
3. Save to: [D:\00_AI_Aplications\ClimateChange\Chapter_01\arcgis_pro\](#)
4. This creates [GeoCascade_Ch01.aprx](#) and a default geodatabase [GeoCascade_Ch01.gdb](#)

Set the coordinate system

For display (the Map):

1. **Map > Properties > Coordinate Systems**
2. Set to: **WGS 1984** (EPSG:4326) for geographic overview

For analysis (measuring distances and areas):

1. In geoprocessing tools, always choose **WGS 1984 UTM Zone 19S** (EPSG:32719)
2. This avoids the ~40% distance error of Web Mercator (EPSG:3857) at latitude 51°S

[!WARNING] **NEVER use EPSG:3857 (Web Mercator) for distance or area calculations at 51°S.** Web Mercator introduces ~40% linear distortion at Patagonian latitudes. Always use a local UTM projection for any measurement-based analysis.

Step 2: Import Chapter 1 Data

Option A: Run the arcpy script (recommended for repeatability)

1. **Analysis > Python Notebook → New Notebook**
2. Paste the content of `01_arcpy_import_imagery.py` into a cell
3. Run (Shift + Enter)

Option B: Manual import (ArcGIS Pro GUI)

Import Copernicus DEM tiles:

1. **Insert > Add Data > Folder Connection** → navigate to `Chapter_01/data/raw/`
2. Expand `dem_*/` folders
3. Drag each `copernicus_dem_30m.tif` to the Map
4. **Data Management > Raster > Mosaic to New Raster:**
 - Input rasters: select all 4 DEM tiles
 - Output: `GeoCascade_Ch01.gdb\CopernicusDEM_Mosaic`
 - Pixel type: **32-bit float**
 - Number of bands: **1**
 - Mosaic operator: **Mean**

Import Sentinel-2 composite:

1. Navigate to the Sentinel-2 scene folder: `sentinel2_l2a_*/`
2. Select `B02.tif`, `B03.tif`, `B04.tif`, `B08.tif`
3. **Data Management > Raster > Composite Bands:**
 - Input rasters: B02, B03, B04, B08 (order matters for band assignment)
 - Output: `GeoCascade_Ch01.gdb\Sentinel2_Composite`

Import CHIRPS climatology:

1. **Insert > Add Data > Raster Dataset**
2. Navigate to: `Chapter_01/data/processed/real_data/chirps_mean_annual_precip.tif`
3. The TIF opens directly with automatic stretching

Import RGI glacier outlines:

1. **Insert > Add Data > File Geodatabase or GeoPackage Layer**
2. Navigate to: `Chapter_01/data/raw/real_data/rgi70_patagonia_glaciers.gpkg`
3. Select the polygon layer and add to Map

[!TIP] All GeoTIFFs produced by the Python pipeline include embedded CRS metadata. ArcGIS Pro reads EPSG codes automatically — no manual projection needed.

Step 3: Create Thematic Maps

Temperature trend map

1. Click the **temperature_surface** layer in the Contents pane
2. **Appearance > Symbology** (or right-click → Symbology)
3. **Primary Symbology** → **Classified**

4. **Method:** Natural Breaks (Jenks), 5 classes
5. **Color scheme:** Search for "Red-Blue Diverging" → select Blue-to-Red
6. **Invert** if needed so Blue = cold, Red = warm
7. **Label** units as **deg C** in the legend

Precipitation map (CHIRPS)

1. Click the **CHIRPS_MeanAnnualPrecip** layer
2. Symbology → **Classified** → Natural Breaks, 5 classes
3. Color scheme: **Yellow to Blue** (Yellow = dry, Blue = wet)
4. Format labels as integers (e.g., **1200 mm**)

DEM + Hillshade

1. First, create the hillshade:
 - **Analysis > Tools > Hillshade** (in 3D Analyst toolbox)
 - Input: **CopernicusDEM_Mosaic**
 - Azimuth: 315 (northwest), Altitude: 45
 - Output: **CopernicusDEM_Hillshade**
2. In Contents pane, drag Hillshade **below** the DEM mosaic layer
3. Set DEM mosaic to **30% transparency** (Appearance tab)

Glacier outlines (RGI)

1. Click **RGI_Glaciers** layer
2. Symbology → **Single Symbol**
3. Symbol type: **Simple Fill**
 - Fill color: **Cyan** (R:0, G:188, B:212), 35% transparency
 - Outline color: **Dark Blue** (R:13, G:71, B:161), width: 1.5 pt
4. Add labels:
 - Right-click layer → **Label Features**
 - Label field: **glac_name**
 - Filter: only label glaciers > 5 km² area

False color Sentinel-2 composite

1. Click **Sentinel2_Composite** layer
2. Symbology → **Stretch**
3. **Band combination:** R=Band 4 (NIR/B08), G=Band 3 (Red/B04), B=Band 2 (Green/B03)
 - This creates a **False Color** view where vegetation = bright red, water = dark, ice/snow = bright cyan
4. **Stretch type:** Percent Clip (2%)

[!NOTE] In standard False Color, vegetation appears **red** because plant leaves strongly reflect Near-Infrared (B08) which is mapped to the red channel. Healthy dense vegetation = bright red, stressed or sparse = pink/brown.

Step 4: Build the Professional Layout

Create a new layout

- 1. **Insert > New Layout > A3 Landscape** (42 × 29.7 cm)
- 2. The layout canvas opens in a new tab

Add 4 map frames

- 1. **Insert > Map Frame** → draw a frame in the top-left quadrant
- 2. Repeat for top-right, bottom-left, bottom-right
- 3. Each frame shows the same map — assign different layer combinations

Suggested panel content:

Panel	Position	Layers visible
Sentinel-2 False Color	Top Left	S2 Composite + Glacier outlines
CHIRPS Precipitation	Top Right	CHIRPS classified + Admin boundaries
Temperature Trend	Bottom Left	Temperature surface + Hillshade base
Glaciers on DEM	Bottom Right	RGI glaciers + DEM hillshade + labels

- 4. For each frame: right-click → **Activate** to pan/zoom to the study area
- 5. Set the same extent in all panels: **BBOX -73.5, -51.5 to -72.5, -50.5**

Add cartographic elements

- 1. **North arrow:** Insert > North Arrow → place in one panel corner
- 2. **Scale bar:** Insert > Scale Bar → choose "Scale Line 1", set units to km
- 3. **Legend:** Insert > Legend → select layers to include
- 4. **Title text:** Insert > Straight Text →

GeoCascade - Torres del Paine Climate Change Analysis

- 5. **Subtitle:** Insert > Straight Text →

Patagonia, Chile | 1993-2024 | ERA5-Land + Sentinel-2 + CHIRPS + RGI 7.0

- 6. **Data sources** (small text, bottom of page):

ERA5-Land: ECMWF/Open-Meteo | CHIRPS: UCSB CHG | Imagery: Planetary Computer
Sentinel-2 L2A: ESA | Landsat: USGS | DEM: Copernicus/ESA | Glaciers: RGI
v7.0

Step 5: Export the Layout

From ArcGIS Pro GUI

1. **Share > Export Layout**
2. **PDF**: resolution 300 DPI, raster quality = Normal, vector quality = Best
3. **PNG**: resolution 200 DPI
4. Save to `Chapter_01/data/processed/`

From arcpy (automated)

```
# Run inside ArcGIS Pro Notebook
exec(open(r'D:\00_AI_Aplications\ClimateChange\Chapter_01\arcgis_pro\03_arcpy_layo
ut_export.py').read())
```

Step 6: Running arcpy Scripts from Notebook

Open the Python Notebook

1. **Analysis > Python Notebook** (in the Analysis ribbon)
2. Click **New** to create a blank notebook

Run a script

Option A — Run file directly:

```
exec(open(r'D:\00_AI_Aplications\ClimateChange\Chapter_01\arcgis_pro\01_arcpy_impo
rt_imagery.py').read())
```

Option B — Paste script into cell:

- Copy the script content into a notebook cell
- Click Run (Shift + Enter)

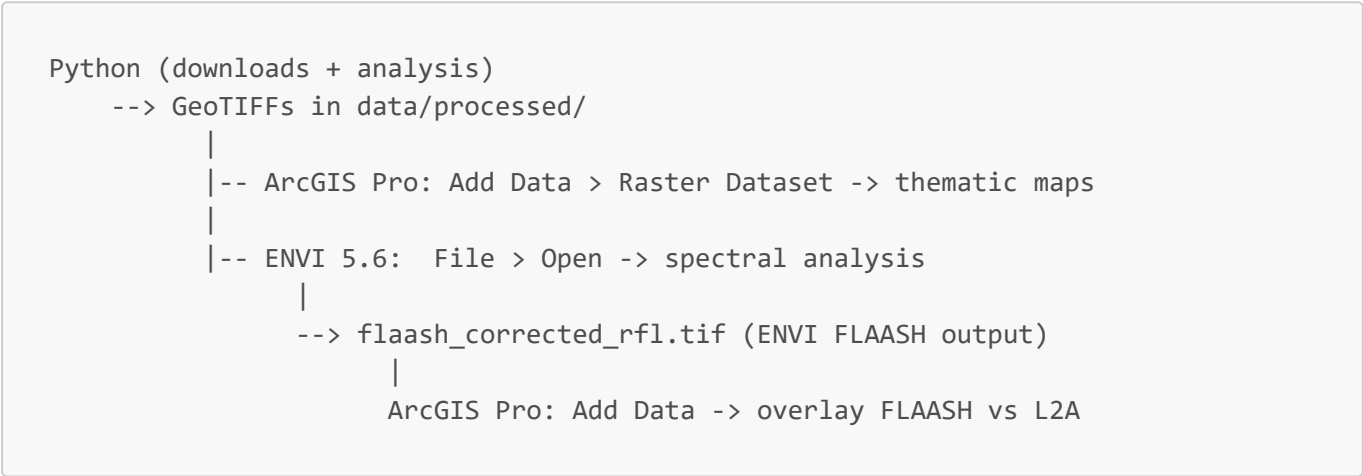
[!IMPORTANT] The ArcGIS Pro Python environment already has `arcpy` installed. **Do NOT install arcpy into `geocascade_env`** — `arcpy` is not available as a standalone package. It only works inside ArcGIS Pro's Python.

[!TIP] You can call the regular GeoCascade Python scripts from ArcGIS Pro Notebook using `subprocess`:

```
import subprocess
subprocess.run([r"C:\path\to\geocascade_env\python.exe",
               r"D:\...\Chapter_01\04_climate_trend_analysis.py"])
```

ArcGIS Pro + ENVI Integration

The workflows connect at the GeoTIFF level:



Files in This Folder

File	Purpose
README_ARCGISPRO.md	This guide
01_arcpy_import_imagery.py	Mosaic DEM, composite S2, import all layers
02_arcpy_climate_maps.py	Apply symbology to all Chapter 1 layers
03_arcpy_layout_export.py	Build A3 layout + export PDF/PNG